

[SECTION-B]
Subject: Digital System Design (BET-C-306)
1st Sessional Exam (CSE 2nd Year)

Time: 1 Hour

SECTION A

M.M.20
(2x5)

Note: Attempt any two question from Section A and one question from Section B.

Q1. A logical expression in the canonical SOP form is as follows:

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}C$$

Minimize it using the K-map technique.

Q2. What is the physical significance of K-map? Using K-map realize the following expression using minimum number of gates

$$Y = \sum m(1, 3, 5, 9, 11, 13)$$

Q3. (a) Differentiate prime implicant and essential prime implicant with example.

(b) Convert the $Y = AB + AC + BC$ into standard SOP form.

Q4. Prove the following Boolean expressions

(a) $A + AB = A$

(b) $A + \bar{A}B = A + B$

SECTION B

(1x10)

Q5. (a) Convert binary 10010 to gray code.

(b) Compute 9's complement of 678

(c) State De-Morgan's Theorem.

Q6. What do you understand by universal gate? Realize NOT, AND and OR gates with the help of universal NAND gate.

Digital System Design (BET-C306)

1st Sessional Branch-CSE III sem Sec-C (faculty Name-Dr. Ashish Nainwal)

Time: 1 Hr

M.M. 20

Note: attempt any two from section A and one from section B.

Section A (6*2=12)

1. (i) Convert $(62)_{10}$ to $(\dots)_2$.
(ii) Convert $(A24)_{16}$ to $(\dots)_8$.
2. (i) Add $+40.75$ to -38.75 using the 12 bit 2's compliments.
(ii) Reduce the expression using Boolean algebra rules. $F = (B+BC)(B+B'C)(B+D)$
3. (i) Perform $254 - 198$ in BCD code.
(ii) Perform $124 + 189$ in XS-3 code.

Section B (8*1=8)

1. Reduce following expression using K-map
(i) $F(A,B,C,D) = \sum m(0,1,2,3,4,5) + d(10,11,12,13,14,15)$.
(ii) $f(A,B,C) = AB' + AB + C + ABC$
2. Reduce following expression using tabular method and implement reduced function using NAND logic.
 $F(A,B,C,D) = \sum m(2,3,8,12,13) + d(10,14)$

Digital System Design (BET-C306)

2nd Sessional Branch-CSE III sem Sec-C (faculty Name-Dr. Ashish Nainwal)

Time: 1Hr

Note: attempt any two from section A and one from section B.

M.M. 20

Section A (6*2=12)

1. Design a half and full adder circuit using decoder.
2. Implement the following Boolean function using 8:1 MUX.
 $F(A,B,C,D)=ABC'+A'BC+AD+ABCD$
3. Convert SR flip flop into JK flip flop.

Section B (8*1=8)

1. Explain 4-bit Parallel adder. Design BCD adder using 74LS83 IC.
2. Design 4 bit binary to gray code converter.